



GATE ME 90 Days Challenge

Day 6/90 · Practice Set with Detailed Solutions

SUBJECT

Weekly Revision

TODAY'S TOPIC

Week 1 Revision

COUNTDOWN

105 days to GATE

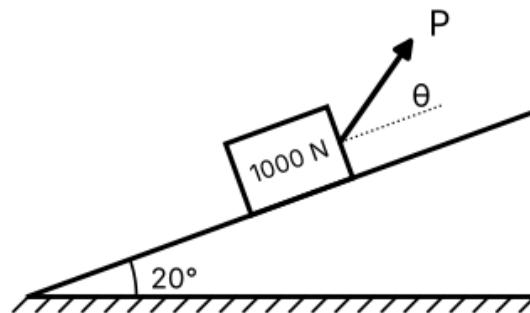
How to use this set: 15 questions (8 MCQ, 7 NAT) on all five topics of Week 1: statics, trusses and friction; kinematics and dynamics; linear algebra; impulse, energy and impact; stress, strain and Mohr's circle (3 questions each). Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 (MCQ)

A 1000 N block is to be pulled up a rough plane inclined at 20° ($\mu = 0.25$) by a force P inclined at θ to the plane, as shown. The least value of P that can just start the block up the plane is



(A) 576.9 N

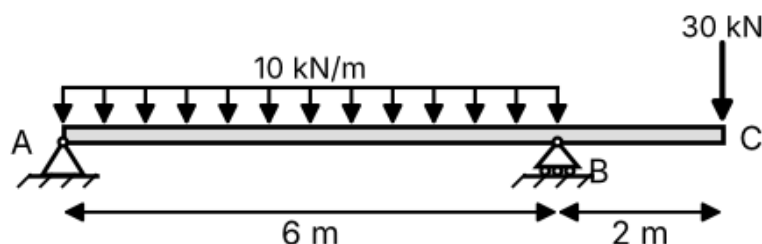
(B) 559.7 N

(C) 675.4 N

(D) 342.0 N

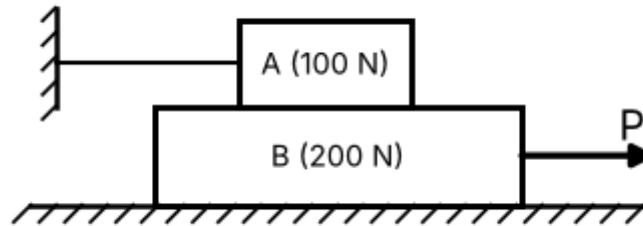
Q2 (NAT)

The beam ABC shown is supported by a pin at A and a roller at B. It carries a UDL of 10 kN/m over AB and a point load of 30 kN at the free end C. The reaction at A (in kN, upward) is _____



Q3 MCQ

Block A (100 N) rests on block B (200 N), which rests on the floor. A is tied to the wall by a horizontal string, as shown. $\mu = 0.3$ at both contact surfaces. The horizontal force P needed to just move B is



- (A) 90 N
(C) 120 N

- (B) 60 N
(D) 150 N

Q4 NAT

A ball is thrown horizontally at 20 m/s from the top of a 45 m high cliff. Its speed just before hitting the level ground below (in m/s) is _____ ($g = 9.81 \text{ m/s}^2$)

Q5 MCQ

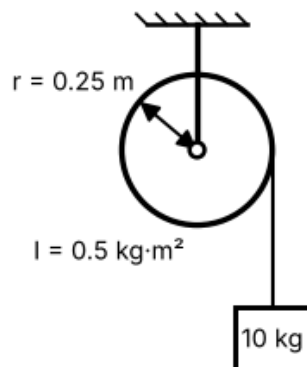
Rain falls vertically at 12 m/s. A man walks on level ground at 5 m/s. To protect himself best, he should hold his umbrella tilted

- (A) forward, at 22.6° to the vertical
(C) forward, at 67.4° to the vertical

- (B) backward, at 22.6° to the vertical
(D) vertically

Q6 NAT

A 10 kg mass hangs from a light rope wound round a drum of radius 0.25 m, which turns on a frictionless fixed axle (moment of inertia of the drum $I = 0.5 \text{ kg}\cdot\text{m}^2$), as shown. The acceleration of the mass when released (in m/s^2) is _____ ($g = 9.81 \text{ m/s}^2$)

**Q7** MCQ

The matrix $\begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & k \\ 3 & 6 & 9 \end{vmatrix}$ has rank 1 when

- (A) $k = 3$
(C) $k = 0$

- (B) $k = 9$
(D) $k = 6$

Q8 (NAT)

For $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$, the value of $\det(A^3)$ is _____

Q9 (MCQ)

If Q is a real orthogonal matrix, which of the following is NOT necessarily true?

- (A) $Q^{-1} = Q^T$ (B) $\det Q = \pm 1$
(C) All eigenvalues of Q are real (D) Every eigenvalue has modulus 1
-

Q10 (NAT)

A car moving at 72 km/h on a level road is braked to rest with all wheels locked. If $\mu = 0.6$ between the tyres and the road, the stopping distance (in m) is _____ ($g = 9.81 \text{ m/s}^2$)

Q11 (MCQ)

A 3 kg ball moving at 4 m/s makes a direct central impact with a 1 kg ball at rest. If $e = 0.8$, the velocities of the 3 kg and 1 kg balls after impact are

- (A) 2 m/s and 6 m/s (B) 2.2 m/s and 5.4 m/s
(C) 3 m/s and 3 m/s (D) 1.8 m/s and 6.6 m/s
-

Q12 (NAT)

A lift of total mass 800 kg is raised at a steady 1.5 m/s. If the efficiency of the drive is 80 %, the input power needed (in kW) is _____ ($g = 9.81 \text{ m/s}^2$)

Q13 (MCQ)

For an isotropic material whose bulk modulus K is equal to its Young's modulus E , Poisson's ratio is

- (A) $1/3$ (B) $1/4$
(C) $1/2$ (D) $1/6$
-

Q14 (NAT)

At a point, $\sigma_x = 40 \text{ MPa}$ (tensile), $\sigma_y = 60 \text{ MPa}$ (compressive) and $\tau_{xy} = 40 \text{ MPa}$. The major principal stress (in MPa) is _____

Q15 (MCQ)

A prismatic bar is fixed between two rigid supports and heated uniformly. The thermal stress developed in it is independent of

- (A) the modulus of elasticity (B) the coefficient of thermal expansion
(C) the temperature rise (D) the length of the bar
-

Answer Key

Q	Type	Answer
Q1	MCQ	(B) 559.7 N
Q2	NAT	20
Q3	MCQ	(C) 120 N
Q4	NAT	35.82 (accept 35.6 to 36.0)
Q5	MCQ	(A) forward, at 22.6° to the vertical
Q6	NAT	5.45 (accept 5.40 to 5.50)
Q7	MCQ	(D) $k = 6$
Q8	NAT	512
Q9	MCQ	(C) All eigenvalues of Q are real
Q10	NAT	33.98 (accept 33.7 to 34.3)
Q11	MCQ	(B) 2.2 m/s and 5.4 m/s
Q12	NAT	14.72 (accept 14.6 to 14.8)
Q13	MCQ	(A) $1/3$
Q14	NAT	54.03 (accept 53.6 to 54.5)
Q15	MCQ	(D) the length of the bar

Detailed Solutions

Q1 Answer: (B) 559.7 N

Concept: P is least when it is inclined to the plane at the angle of friction $\phi = \tan^{-1}\mu$; then $P_{\min} = W \sin(\alpha + \phi)$.

$$\phi = \tan^{-1}(0.25) = 14.04^\circ$$

$$P_{\min} = 1000 \sin(20^\circ + 14.04^\circ) = 1000 \sin 34.04^\circ = \mathbf{559.7 \text{ N}}, \text{ at } \theta = 14.04^\circ$$

Check: P parallel to the plane needs $W(\sin \alpha + \mu \cos \alpha) = 576.9 \text{ N}$, which is larger. ✓

Exam tip: "Least force" problems: tilt the force by ϕ . For a horizontal floor the answer reduces to $P_{\min} = W \sin \phi$.

Q2 Answer: 20

Concept: Take moments about the other support to get one reaction directly.

$$\text{Moments about B (clockwise +): } R_A(6) - 60(3) + 30(2) = 0$$

$$R_A = (180 - 60)/6 = \mathbf{20 \text{ kN}}; R_B = 60 + 30 - 20 = 70 \text{ kN}$$

Exam tip: A load on the overhang relieves the far support: here it reduces R_A from 30 kN to 20 kN. (Useful for Day 7: SFD & BMD.)

Q3 Answer: (C) 120 N

Concept: Draw the FBD of B: it is resisted by friction from A on its top face and from the floor on its bottom face.

Top face: $F_1 = \mu N_1 = 0.3 \times 100 = 30 \text{ N}$ (A is held by the string)

Bottom face: $N_2 = 100 + 200 = 300 \text{ N}$, $F_2 = 0.3 \times 300 = 90 \text{ N}$

$P = 30 + 90 = 120 \text{ N}$

Exam tip: If A were free (no string), the blocks would move together and $P = 90 \text{ N}$ only.

Q4 Answer: 35.82 (accept 35.6 to 36.0)

Concept: Horizontal velocity stays constant; vertical velocity grows as in free fall.

$v_y^2 = 2gh = 2 \times 9.81 \times 45 = 882.9 \rightarrow v_y = 29.71 \text{ m/s}$

$v = \sqrt{20^2 + 882.9} = 35.82 \text{ m/s}$, at 56.1° below the horizontal

Exam tip: Energy gives it in one line: $v^2 = u^2 + 2gh$.

Q5 Answer: (A) forward, at 22.6° to the vertical

Concept: The umbrella must face the velocity of rain relative to the man: $v_{\text{rain/man}} = v_{\text{rain}} - v_{\text{man}}$

Relative velocity: 12 m/s down and 5 m/s backwards; magnitude 13 m/s

Angle with the vertical: $\tan^{-1}(5/12) = 22.6^\circ$, so rain appears to come from the front $\rightarrow$ tilt the umbrella **forward**.

Exam tip: $\tan^{-1}(12/5) = 67.4^\circ$ is the angle with the horizontal, a common trap.

Q6 Answer: 5.45 (accept 5.40 to 5.50)

Concept: Mass: $mg - T = ma$. Drum: $Tr = I\alpha$ with $a = \alpha r$.

$T = I\alpha/r^2 = 0.5a/0.0625 = 8a$

$10 \times 9.81 - 8a = 10a \rightarrow a = 98.1/18 = 5.45 \text{ m/s}^2$; $T = 43.6 \text{ N}$

Exam tip: The drum adds an "equivalent mass" I/r^2 ($= 8 \text{ kg}$ here) in the denominator: $a = mg/(m + I/r^2)$.

Q7 Answer: (D) $k = 6$

Concept: Rank 1 means every row is a multiple of the first row.

$R_3 = 3R_1$ already. $R_2 = 2R_1$ needs $k = 2 \times 3 = 6$.

For any other k , rows 1 and 2 are independent and the rank is 2 (it can never be 3, since $R_3 = 3R_1$).

Exam tip: For rank questions, look for proportional rows before expanding determinants.

Q8 Answer: 512

Concept: $\det(A^n) = (\det A)^n$; also $\det A = \text{product of eigenvalues}$.

$\det A = 9 - 1 = 8$ (eigenvalues 4 and 2: $4 \times 2 = 8$ ✓)

$\det(A^3) = 8^3 = 512$

Exam tip: Never multiply the matrix out: work with the determinant or eigenvalues (here $64 \times 8 = 512$).

Q9 Answer: (C) All eigenvalues of Q are real**Concept:** Orthogonal: $Q^T Q = I$.So $Q^{-1} = Q^T$, $\det Q = \pm 1$, and all eigenvalues have modulus 1.But the eigenvalues need not be real: the 2-D rotation matrix $[\cos \theta, -\sin \theta; \sin \theta, \cos \theta]$ has eigenvalues $e^{\pm i\theta}$. Hence **(C)** is not necessarily true.**Exam tip:** Symmetric $\rightarrow$ real eigenvalues; orthogonal $\rightarrow |\lambda| = 1$; skew-symmetric $\rightarrow$ zero or purely imaginary.**Q10 Answer: 33.98 (accept 33.7 to 34.3)****Concept:** Work-energy: friction work = loss of KE; the mass cancels.

$$u = 72 \text{ km/h} = 20 \text{ m/s}$$

$$\mu mg s = \frac{1}{2} mu^2 \rightarrow s = u^2 / (2\mu g) = 400 / (2 \times 0.6 \times 9.81) = \mathbf{33.98 \text{ m}}$$

Exam tip: Stopping distance $\propto u^2$: doubling the speed makes it four times longer.**Q11 Answer: (B) 2.2 m/s and 5.4 m/s****Concept:** Momentum + Newton's law of restitution.

$$3(4) = 3v_1 + v_2 \text{ and } v_2 - v_1 = 0.8 \times 4 = 3.2$$

$$4v_1 = 12 - 3.2 \rightarrow v_1 = \mathbf{2.2 \text{ m/s}}, v_2 = \mathbf{5.4 \text{ m/s}}$$

Check: $e = 1$ gives 2 and 6 m/s; $e = 0$ gives 3 and 3 m/s.**Exam tip:** General results: $v_1 = (m_1 - em_2)u_1 / (m_1 + m_2)$, $v_2 = (1 + e)m_1 u_1 / (m_1 + m_2)$.**Q12 Answer: 14.72 (accept 14.6 to 14.8)****Concept:** At constant speed, the pull equals the weight; power = force $\times$ velocity.

$$\text{Output power} = 800 \times 9.81 \times 1.5 = 11\,772 \text{ W}$$

$$\text{Input} = 11\,772 / 0.8 = 14\,715 \text{ W} = \mathbf{14.72 \text{ kW}}$$

Exam tip: Divide by efficiency for input; multiplying (9.42 kW) is the common slip.**Q13 Answer: (A) 1/3****Concept:** $E = 3K(1 - 2v)$.

$$K = E \rightarrow 1 = 3(1 - 2v) \rightarrow 1 - 2v = 1/3 \rightarrow v = \mathbf{1/3}$$

$$\text{Then } G = E / [2(1 + 1/3)] = 3E/8.$$

Exam tip: Remember the landmarks: $v = 1/3 \rightarrow K = E$; $v = 0.5 \rightarrow K = \infty$; $v = 0 \rightarrow K = E/3$.**Q14 Answer: 54.03 (accept 53.6 to 54.5)****Concept:** $\sigma_1 = C + R$ from Mohr's circle.

$$C = (40 - 60)/2 = -10 \text{ MPa}$$

$$R = \sqrt{(50^2 + 40^2)} = 64.03 \text{ MPa}$$

$$\sigma_1 = -10 + 64.03 = \mathbf{54.03 \text{ MPa}}; \sigma_2 = -74.03 \text{ MPa}$$

Exam tip: Keep signs: $\sigma_y = -60$, so $(\sigma_x - \sigma_y)/2 = 50$, not -10 .

Q15 Answer: (D) the length of the bar**Concept:** $\sigma = E\alpha \Delta T$ for full restraint.

The strain prevented is $\alpha\Delta T$ (a strain, not a length), so the stress depends on E , α and ΔT only.

It is independent of **the length** (and of the cross-sectional area). The force, σA , does depend on the area.

Exam tip: Length matters only when there is a gap or a yielding support: then strain = $(\alpha\Delta T L - \text{gap})/L$.

What next?

Day 6 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124



Website test series



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Next, Day 7/90: Strength of Materials, SFD & BMD. Comment **GATE90** on our reel to get all links. Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.